

Signals & Systems (CSE2090) 2025 Spring

Prerequisite : Engineering Mathematics, Circuit & Lab

An Elective Course for CSE students

(A Mandatory Course for ICE students)

3 Credit-Hour Course

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B.S.	SNU (Seoul National University, Korea) School of Electrical Engineering
M.S.	KAIST (Korea Advanced Inst. of Science and Tech., Korea) School of Electronics and Electrical Engineering (Major : Communications, Digital Signal Processing)
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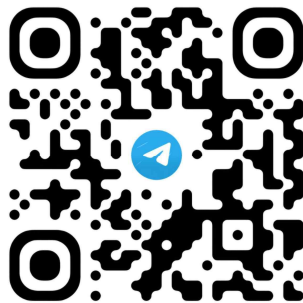
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Textbook:

Alan V. Oppenheim and Alan S. Willsky, Signals and Systems (2nd Ed.),
Prentice-Hall, 1997.

References

1. Simon Haykin, Barry Van Veen, Signals and Systems, Wiley, 2002.
2. McClellan, Schafer and Yoder, DSP First (2nd Ed.), Pearson, 2017.
3. MIT OpenCourseWare(OCW)

https://ocw.mit.edu/search/?d=Electrical%20Engineering%20and%20Computer%20Science&s=department_course_numbers.sort_coursenum

MIT OPEN COURSEWARE
MASSACHUSETTS INSTITUTE OF TECHNOLOGY

6.003 | Fall 2011 | Undergraduate
Signals And Systems

Course Description
6.003 covers the fundamentals of signal and system analysis, focusing on representations of discrete-time and continuous-time signals (singularity functions, complex exponentials and geometrics, Fourier representations, Laplace and Z transforms, sampling) and representations of linear, time-invariant systems ...[Show more](#)

Course Info

INSTRUCTOR
[Prof. Dennis Freeman](#)

DEPARTMENTS
[Electrical Engineering and Computer Science](#)

TOPICS
 Engineering
 Electrical Engineering
 Digital Systems
 Signal Processing

LEARNING RESOURCE TYPES

[Lecture Videos](#)
[Problem Sets with Solutions](#)
[Exams with Solutions](#)
[Online Textbook](#)
[Lecture Notes](#)

6.003
Signals and Systems
modeling — analysis — design

Course logo for 6.003, highlighting the main themes of the course. (Figure by Prof. Dennis Freeman.)

[Download Course](#)

Grading

Midterm Exam	40 %
Final Exam	45 %
Attendance	5 %
Homeworks	10 %

A: $20 \pm 5 \%$ B: $30 \pm 5 \%$ C: $30 \pm 5 \%$ D&F: $20 \pm 5 \%$

Homework Policies

- All homeworks must be done by yourselves.
(Do not copy, please. If copy, you will get ZERO.)
- Upload homeworks on the eclass within the first due dates.
(within the second due dates : 80%)

How to Study

- If you have any question :
 1. Read textbooks and reference books thoroughly FIRST.
 2. THINK, THINK, THINK.
 3. Try to get solutions by yourself.
 4. THINK, THINK, THINK.
 5. Get some information from internet.
 6. THINK, THINK, THINK.
 7. Discuss with your friends.
 8. THINK, THINK, THINK.
 9. Try to get solutions by yourself, again and again.
 10. THINK, THINK, THINK.
 11. Use the Q&A in the e-class or
send me messages using telegram any time.

BEFORE THINKING, DO NOT ASK

MUST change the studying method

(You are not a high school student, but a university student.)

※ Study Group

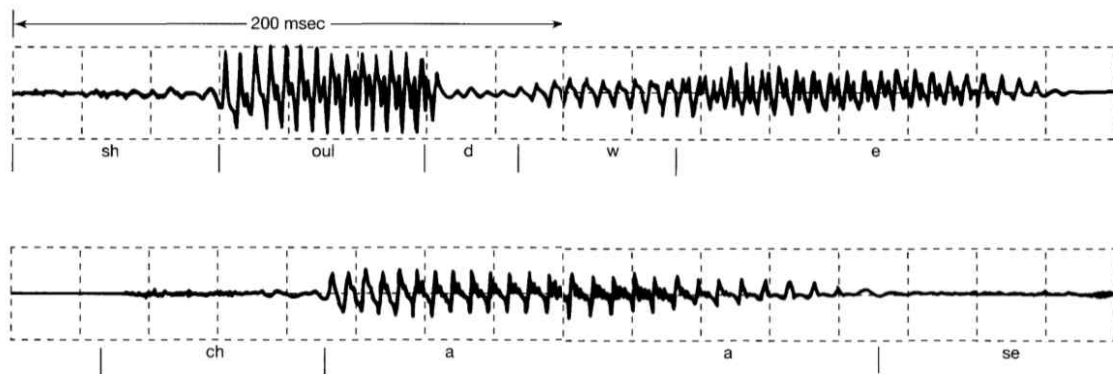
Course Tree of Circuits and Systems

Year	Subject	Remark
Sophomore	Digital Logic Circuits Electric Circuits	
Junior	Electronic Circuits Signals and Systems	ICE
Senior & Graduate School	Digital Signal Processing (DSP Applications) Multimedia Applications Image Processing Speech Signal Processing Digital Communications Mobile Communication Wireless Communication Internet of Things (IoT) Robotics	No in IUT

Applications of Signals and Systems

(Example) A speech signal (1-D signal)

" Should we chase?"



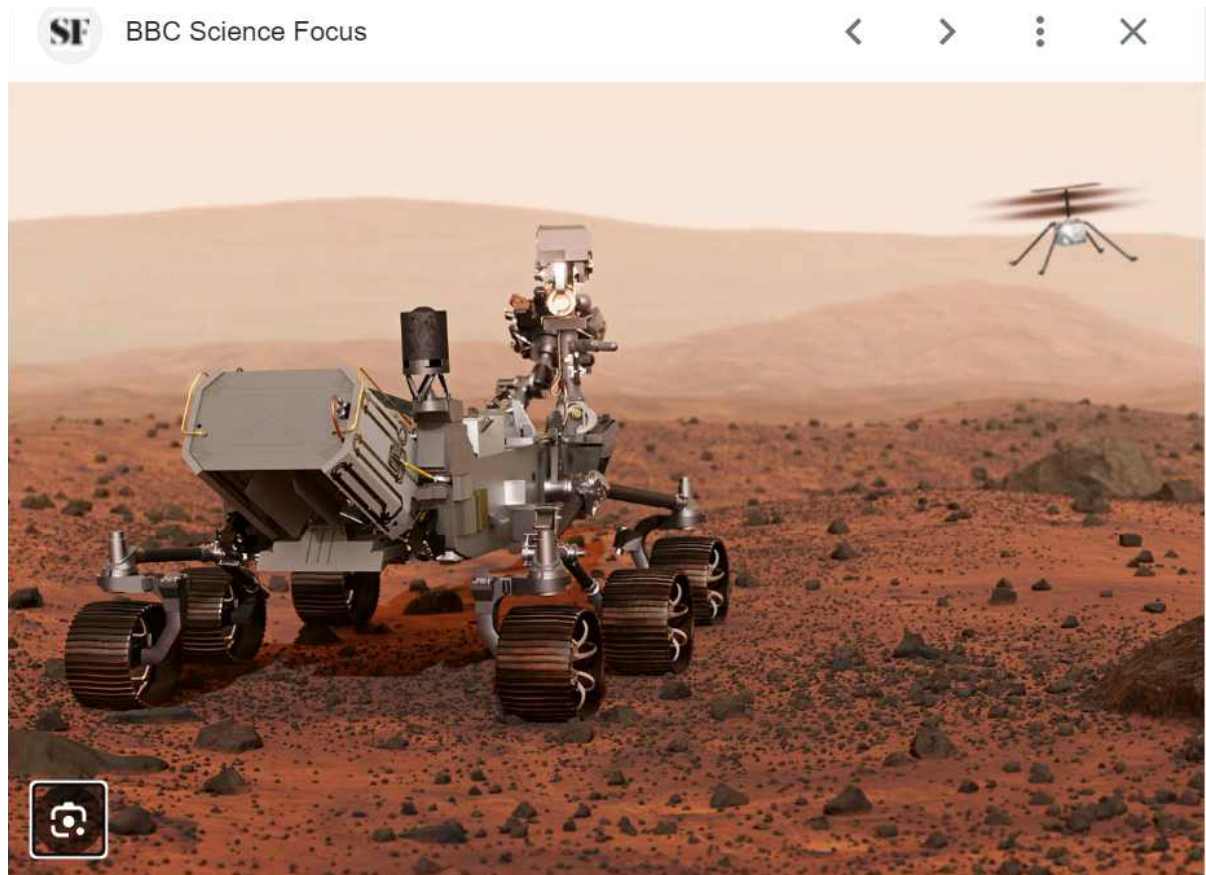
- (1) How can we store the signal?
- (2) How can we make the signal?
- (3) How can we modify the signal?
- (4) How can we analyze the signal?
- (5) How can we recognize the signal?

(Example) A picture signal (2-D signal)



※ Photoshop (2-Dimensional LPF, BPF and HPF)

(Example) NASA's Perseverance Mars rover and helicopter



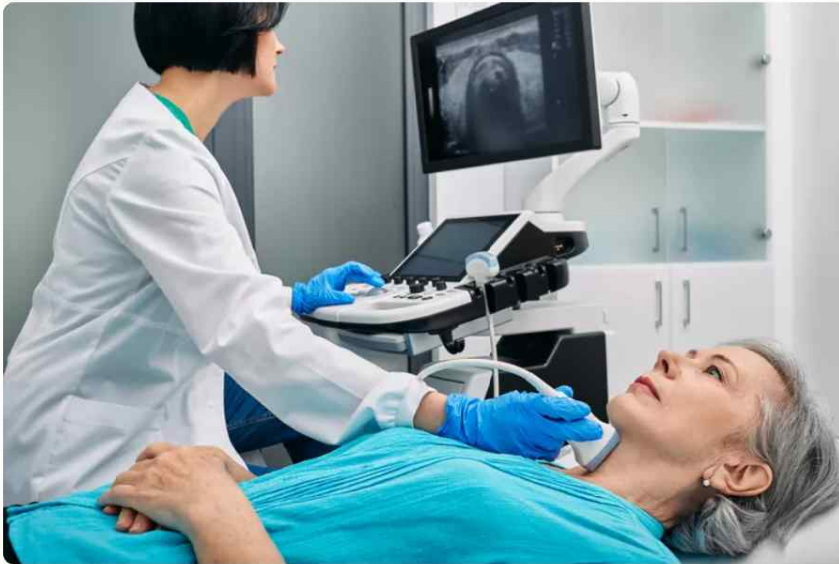
How can we take pictures (that is, make a camera)?

How can we send the image signal from the Mars to the Earth?

How can we remove the noise from the received signal?

How can we remake this analog image from the 1-D received digital signal?

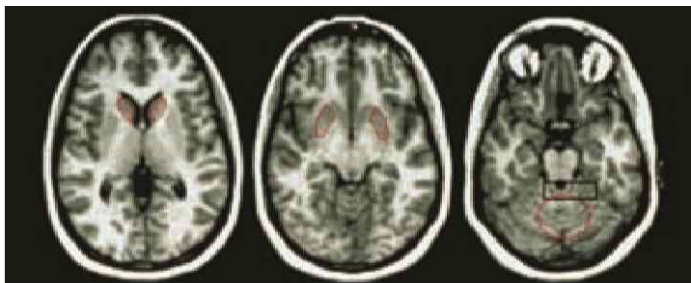
(Example) Ultrasound scanner



How can we see the inside of the body?

(Example) Computer Tomography (CT)

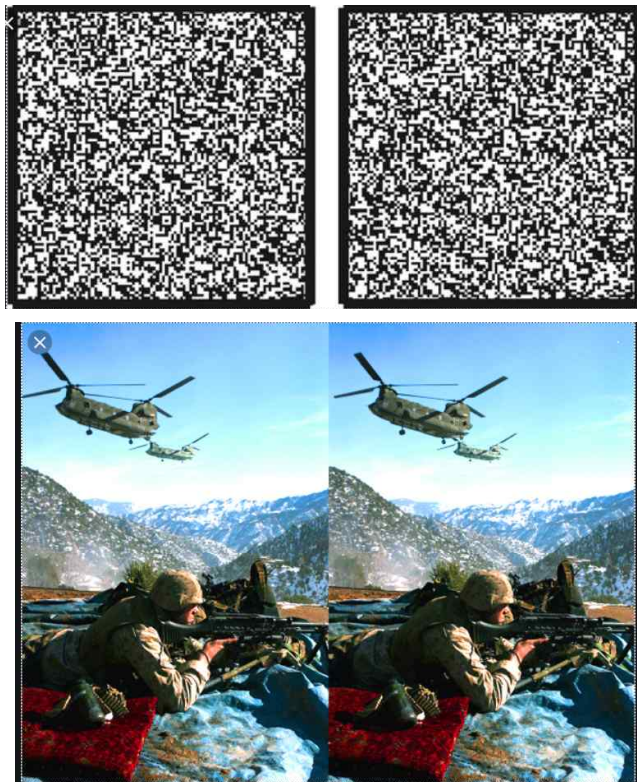
- cross section of a human brain



How can we make these images from data.

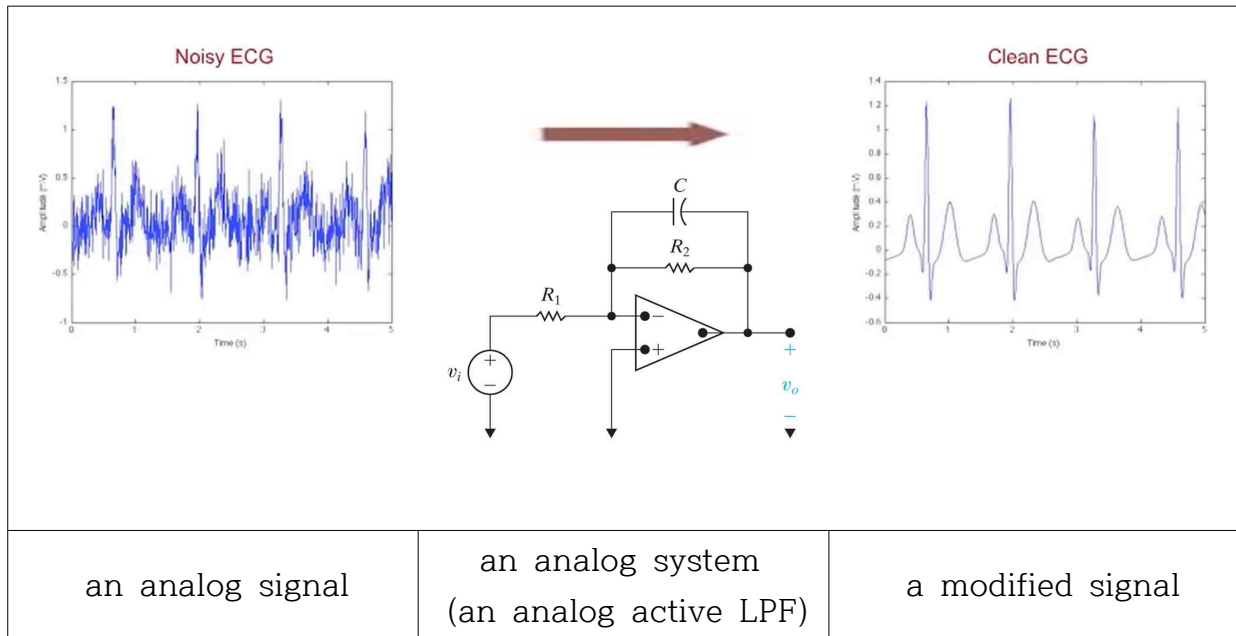
(Fourier Transform - Laplace Transform)

(Example) Stereo Vision (3-D image)



How can we make a 3-D image using 2-D images?

(Example) ECG signal (ElectroCardio Graph) [1-D filtering]



How can we modify (improve) a signal by a system (i.e., filter)?

How can we make a digital low-pass filter for a computer?

How can we modify (improve) an analog/digital signal
by an analog/digital system?

(Example) 2-D filtering

original digital image	filtering	filtered digital image	remark
	low-pass filtering (LPF)		blurring
	high-pass filtering (HPF)		edge detection

How can we modify (improve) a signal by a system?

How can we make a digital LPF/HPF for a computer?

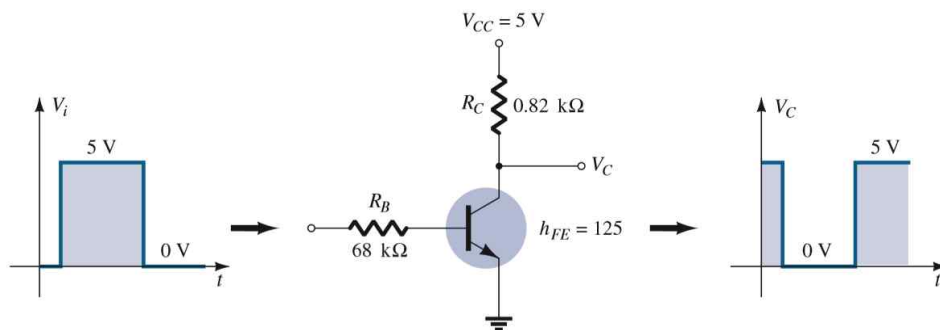
Analog system / Digital system

analog system	hardware	electric circuits electronic circuits	Which is better?
	software	?	
digital system	hardware	digital logic circuits	
	software	programming languages	

* memory, noise, flexibility

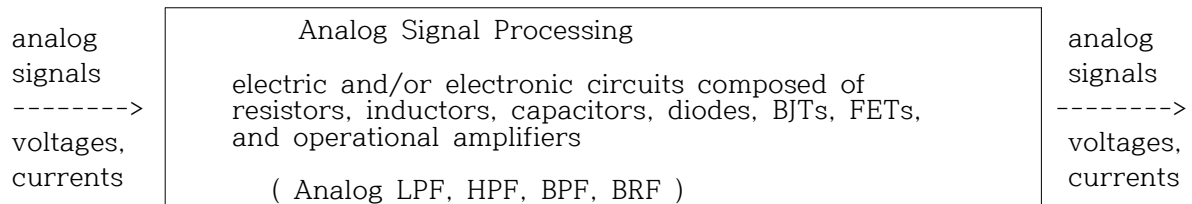
* How can we make digital logic circuits?

(Ex) NOT gate



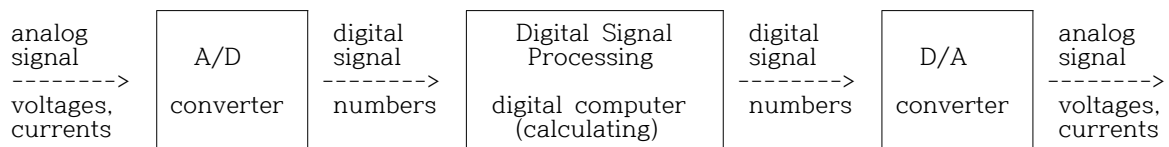
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- What is a signal?
 - What is a system?
 - What is the convolution integral?
 - How can we represent a signal or a system in time domain?
differential equations
 - How can we represent a signal or a system in freq. domain?
Phasor Transform, Laplace Transform

- Continuous Signal / System

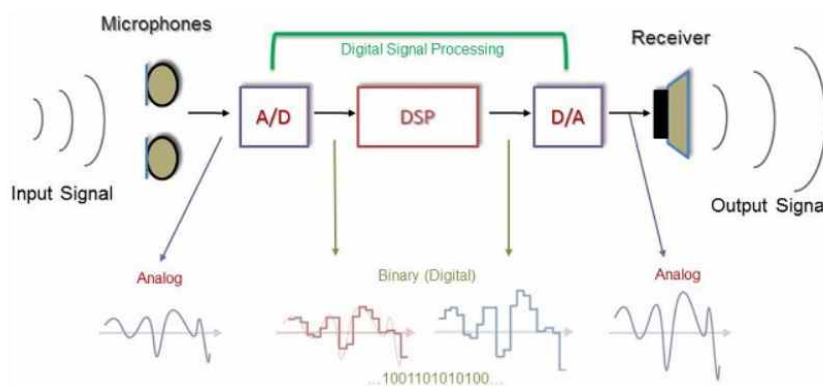


- Digital Signal / System

digital signal : a sequence of numbers



(Example) a digital amplifier



We must use digital computers in these days.

How can we make digital versions of analog filters?

Computer cannot understand analog quantities.

How can we represent analog quantities (time and frequency) for digital computers? [We will study in this class.]

- Why do we use DSP instead of ASP these days?
- Can we use the (.....) for digital systems?
 - convolution integral
 - differential equations
 - phasor transform
 - Laplace transform
- Finally, how can we design digital filters for **digital signal processing**?
 - Can we use analog filter design methods in order to design digital filters?

Digital Signal Processing (DSP)

Signal Processing using DIGITAL COMPUTERS

DSP Advantages

1. Guaranteed accuracy
2. Perfect reproducibility (Noise)
3. No drift in performance with temperature or age
4. Greater reliability, smaller size, lower cost, lower power consumption, higher speed
5. Greater flexibility
6. Memory
7. Superior performance (ex: linear phase response)

DSP Disadvantages

1. Speed and cost
2. Design time
3. Finite wordlength problem

Application areas

1. Image Processing
2. Instrumentation/Control
3. Speech/Audio
4. Military
5. Telecommunications
6. Biomedical